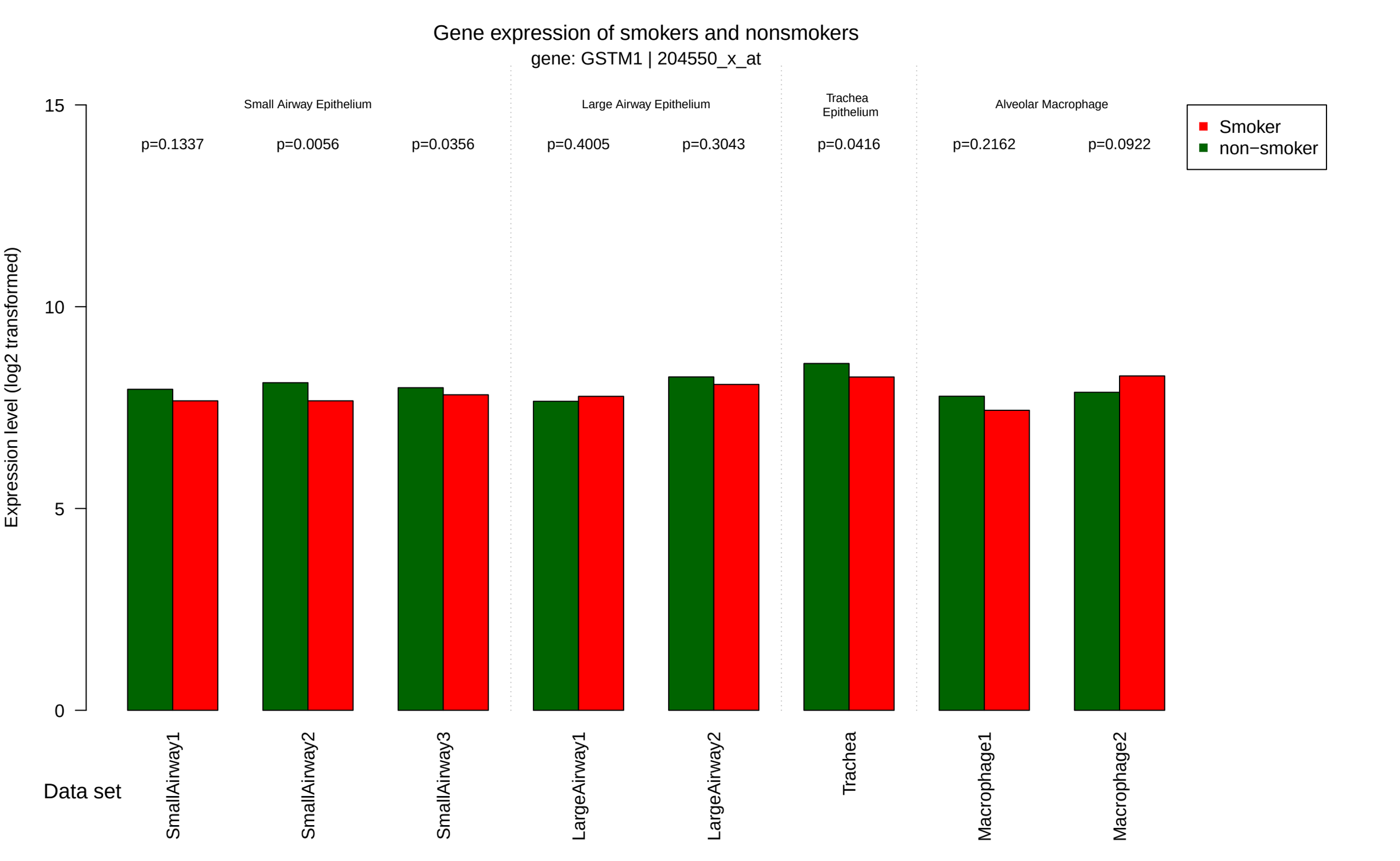
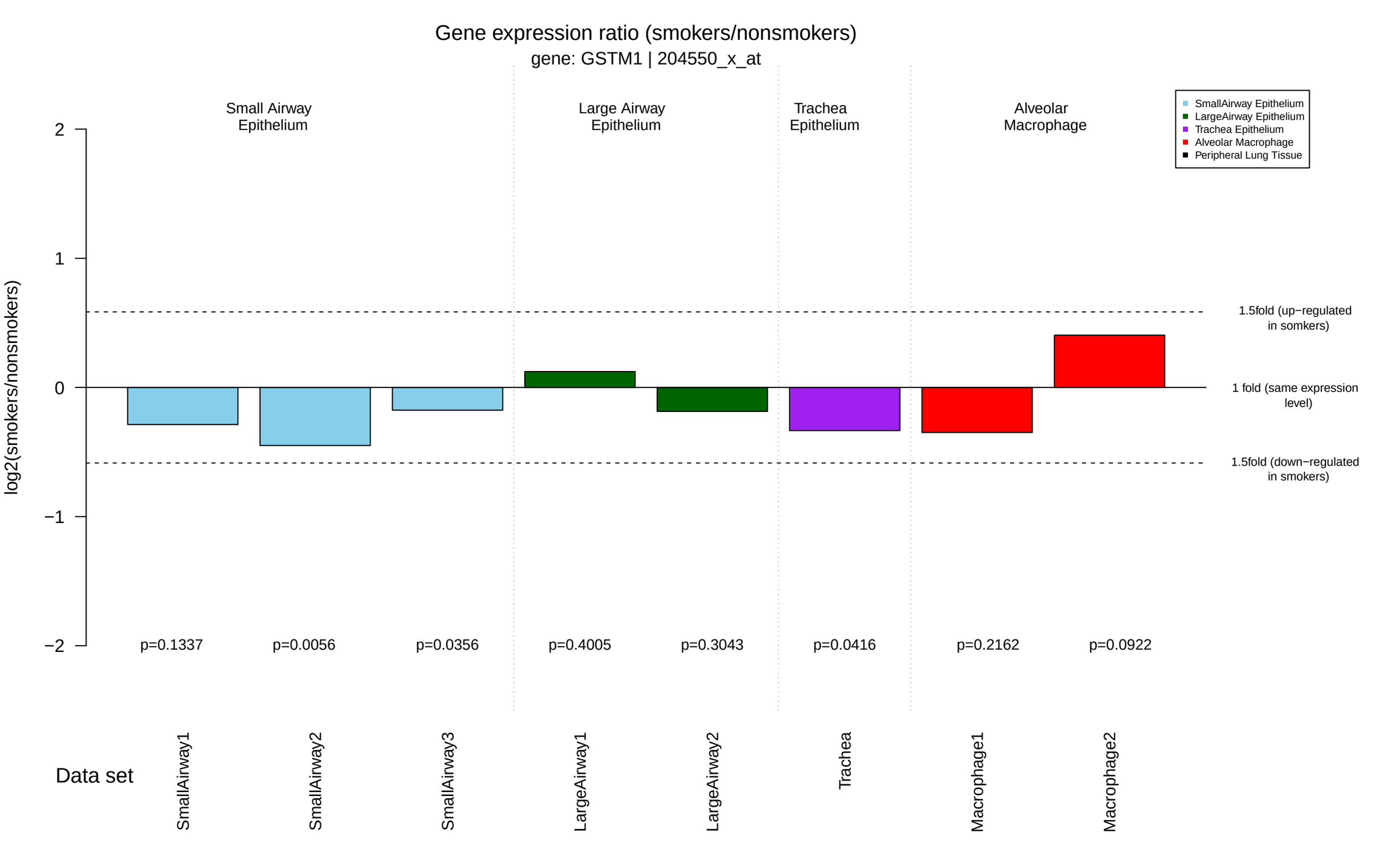
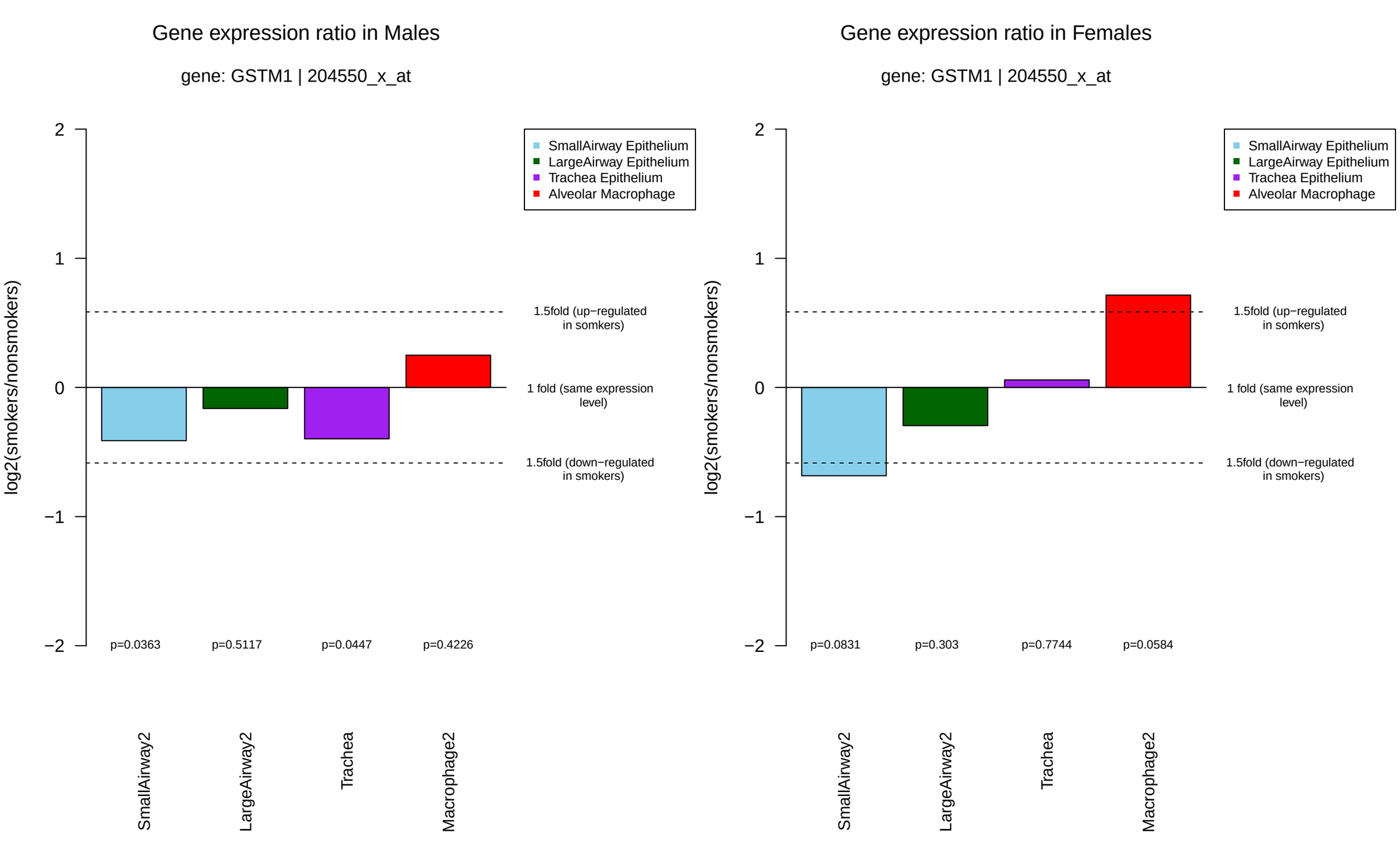
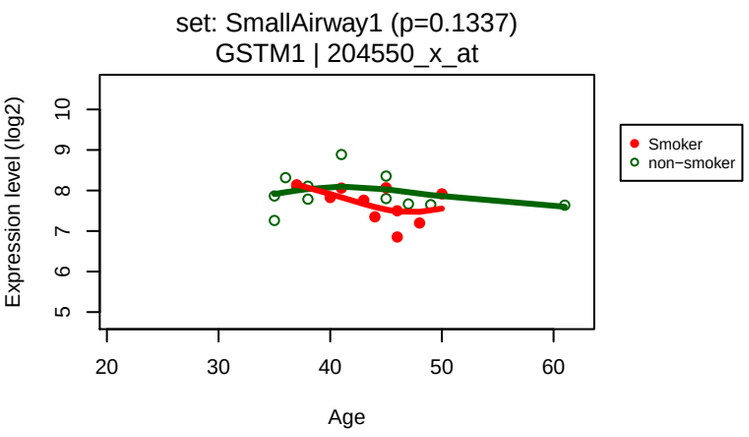
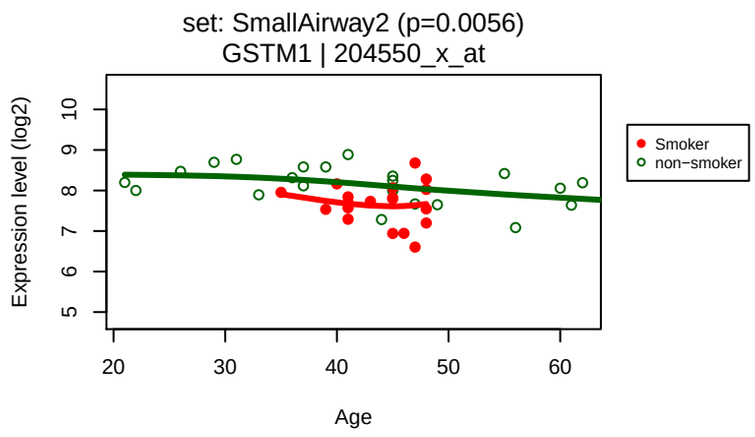
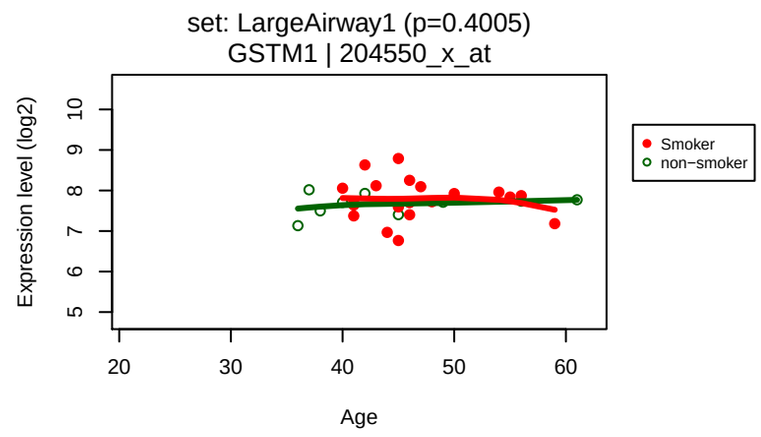

Gene expression of smokers and nonsmokers
gene: GSTM1 | 204550_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1337
p=0.0056
p=0.0356
p=0.4005
p=0.3043
p=0.0416
p=0.2162
p=0.0922
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

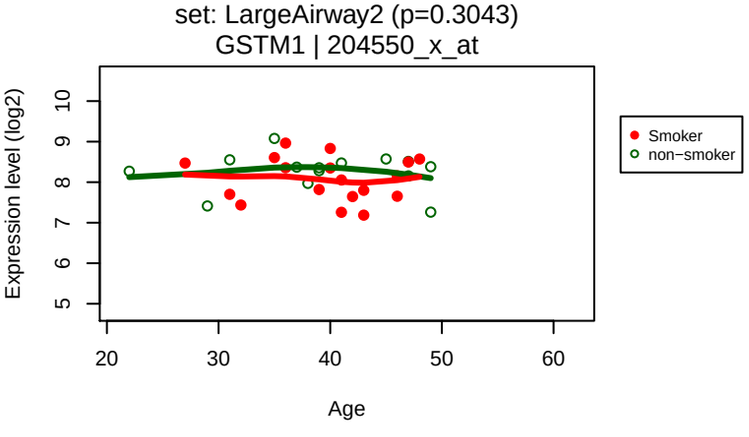

Gene expression ratio (smokers/nonsmokers)
gene: GSTM1 | 204550_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1337
p=0.0056
p=0.0356
p=0.4005
p=0.3043
p=0.0416
p=0.2162
p=0.0922
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

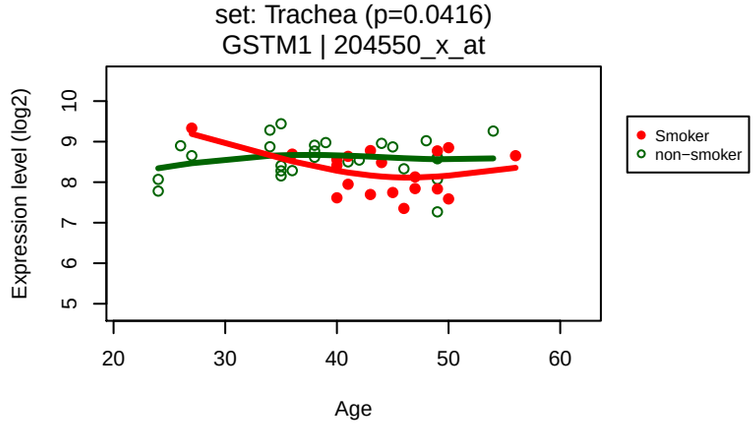

Gene expression ratio in Males
gene: GSTM1 | 204550_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0363
p=0.5117
p=0.0447
p=0.4226
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTM1 | 204550_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0831
p=0.303
p=0.7744
p=0.0584
SmallAirway2
LargeAirway2
Trachea
Macrophage2

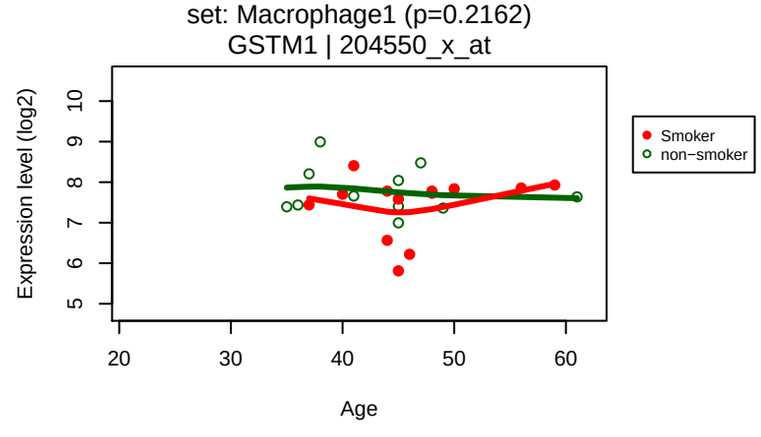

set: SmallAirway1 (p=0.1337)
GSTM1 | 204550_x_at
Expression level (log2)
Age
Smoker
non−smoker

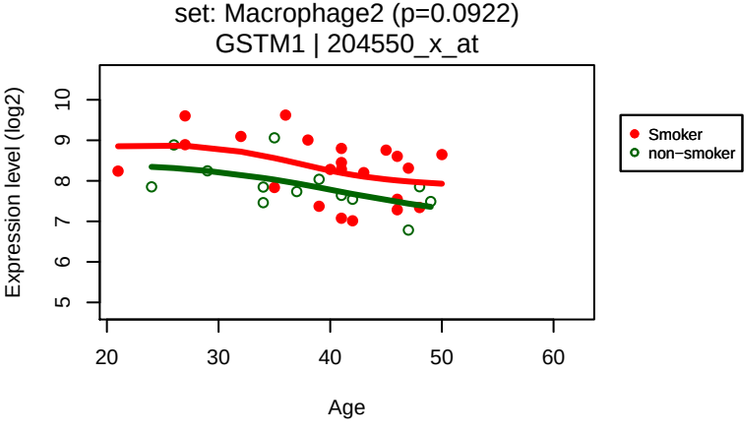

set: SmallAirway2 (p=0.0056)
GSTM1 | 204550_x_at
Expression level (log2)
Age
Smoker
non−smoker

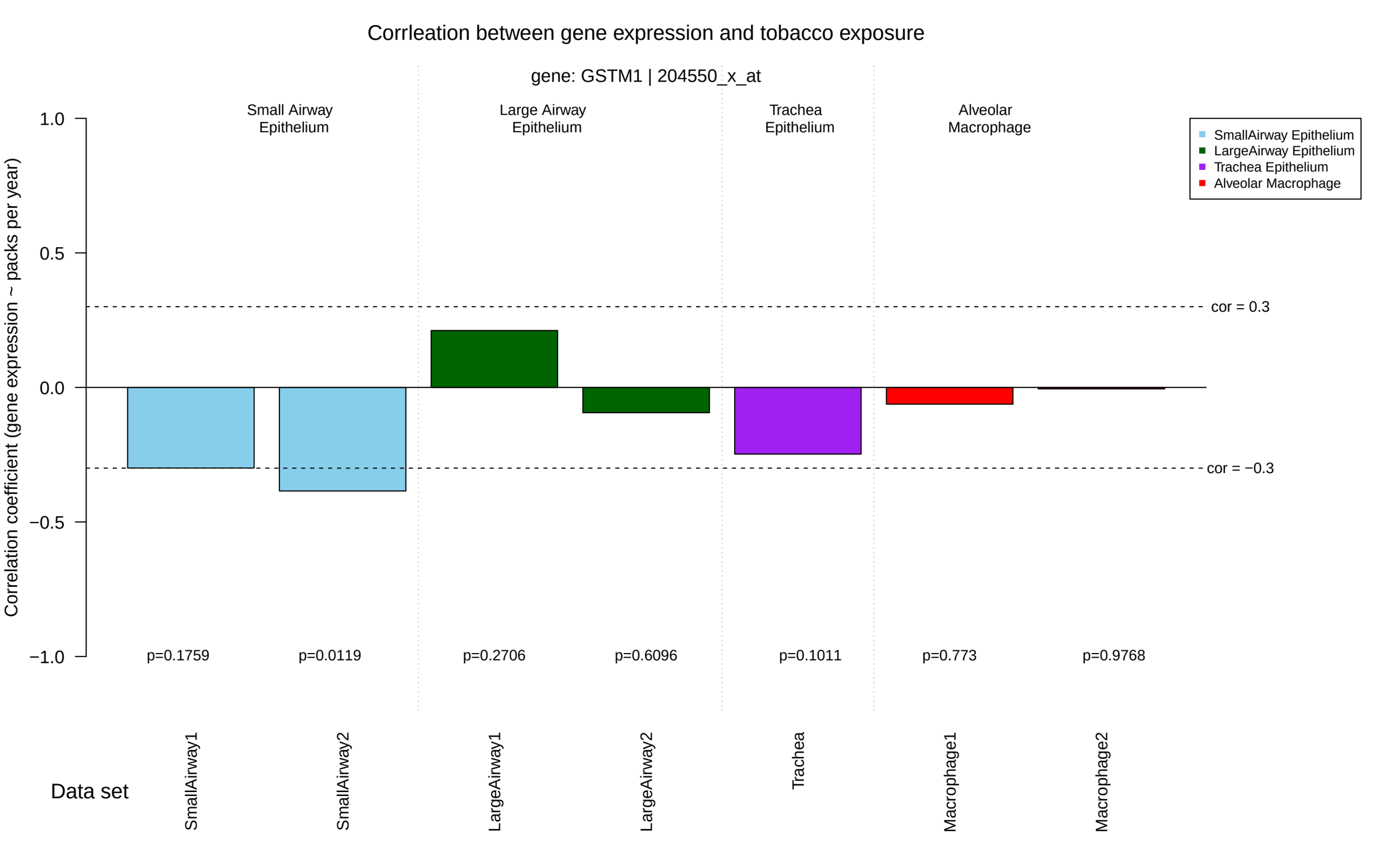

set: LargeAirway1 (p=0.4005)
GSTM1 | 204550_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3043)
GSTM1 | 204550_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0416)
GSTM1 | 204550_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2162)
GSTM1 | 204550_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0922)
GSTM1 | 204550_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTM1 | 204550_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1759
p=0.0119
p=0.2706
p=0.6096
p=0.1011
p=0.773
p=0.9768
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2